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Quantum Electrodynamics

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Contents

Introduction

§ 1. Uncertainty Relations in the Relativistic Domain

The quantum theory presented in Vol. III of this course is non-relativistic in its very substance, and it fails for phenomena in which motion occurs at speeds no longer small next to the speed of light. One might at first imagine that a relativistic theory could be reached by generalizing the formalism of non-relativistic quantum mechanics in a more or less straightforward manner. A closer look shows, however, that a logically complete relativistic theory cannot be built without invoking fresh physical principles.

We begin by recalling several physical notions on which non-relativistic quantum mechanics rests (Vol. III, § 1). A fundamental part was played there, as we saw, by the concept of measurement — the process in which a quantum system interacts with a "classical object" (an "apparatus") and thereby comes to possess definite values of particular dynamical variables (coordinates, velocities, and so forth). We saw, too, that quantum mechanics places severe limits on the extent to which different dynamical variables of an electron[1] can exist together. Thus the uncertainties Δq and Δp with which coordinate and momentum can coexist obey the relation $\Delta q \Delta p \sim \hbar$;[2] the more precisely either of these quantities is measured, the less precisely the other can be known at the same moment.

What matters, however, is that any single dynamical variable of the electron, on its own, admitted measurement of unlimited precision, and within a time interval that could be taken as short as desired. On this circumstance the whole of non-relativistic quantum mechanics depends in a fundamental way. It alone makes it possible to introduce the wave function, the central object of the theory's formalism. The physical content of the wave function $\psi(q)$ is, after all, that its squared modulus gives the probability that a measurement performed at a given instant will yield one or another value of the electron's coordinate. Evidently the very notion of such a probability presupposes that a coordinate measurement of unlimited precision and speed can be realized in principle; failing that, the notion would be empty of content and would forfeit its physical meaning.

The existence of a limiting speed (the speed of light c) leads to new limitations, of a fundamental kind, on the possibilities of measuring various physical quantities (*L. D. Landau, R. Peierls*, 1930).

In Vol. III, § 44 the relation

$$(v' - v)\,\Delta p\,\Delta t \sim \hbar \tag{1.1}$$

was obtained; it relates the uncertainty Δp of a momentum measurement on the electron to the length Δt of the measuring process itself, v and v' denoting the electron's velocity before and after that process. It follows that momentum can be measured accurately within a short time (small Δp together with small Δt) only if the measurement itself

[1] As in Vol. III, § 1, we say "electron" for brevity, understanding by it any quantum system.

[2] Ordinary units are used in this section.

alters the velocity strongly enough. In the non-relativistic theory this expresses the fact that a momentum measurement repeated after a short interval will not reproduce its result, but it places no restriction of principle on how accurate a single measurement of the momentum may be, the difference $v' - v$ being capable of unlimited increase.

The presence of a limiting speed, however, changes the state of affairs radically. The difference $v' - v$, like the velocities themselves, can now not exceed c (more precisely, $2c$). Replacing $v' - v$ in (1.1) by c, we obtain the relation

$$\Delta p \, \Delta t \sim \hbar / c, \tag{1.2}$$

which determines the best accuracy of momentum measurement attainable in principle for a given duration Δt of the measurement. Thus in the relativistic theory an arbitrarily accurate and arbitrarily rapid measurement of the momentum turns out to be impossible in principle. An exact measurement of the momentum ($\Delta p \to 0$) is possible only in the limit of an infinitely long duration of the measurement.

One has reason to expect that the measurability of the electron's coordinate as such is likewise affected. Within the mathematical formalism of the theory this appears as a conflict between an exact coordinate measurement and the statement that a free particle's energy is positive. As we shall see, the full set of eigenfunctions of a free particle's relativistic wave equation contains, besides the solutions whose time dependence is the "proper" one, solutions of "negative frequency" as well. These will in general appear also in the expansion of a wave packet describing an electron confined to a small region of space.

As will be shown, the "negative-frequency" wave functions are tied to the existence of antiparticles — the positrons. Their presence in the expansion of the wave packet reflects the creation of electron–positron pairs, unavoidable in the general case, during a measurement of the electron's coordinates. Since the emergence of these new particles cannot be controlled by the measuring process itself, coordinate measurement for the electron is drained of meaning.

In the rest frame of the electron the minimum error in a measurement of its coordinates is

$$\Delta q \sim \hbar / mc. \tag{1.3}$$

This value (the only one that dimensional considerations would permit) answers to a momentum uncertainty $\Delta p \sim mc$, which corresponds in turn to the minimum threshold energy needed to create a pair.

In a reference frame where the electron moves with energy ε, (1.3) is replaced by

$$\Delta q \sim c\hbar / \varepsilon. \tag{1.4}$$

In the extreme ultra-relativistic limit, in particular, energy and momentum are related by $\varepsilon \approx cp$, so that

$$\Delta q \sim \hbar / p, \tag{1.5}$$

the error Δq then coinciding with the particle's de Broglie wavelength.[3]

[3]What is meant here are measurements any outcome of which permits an inference about the state of

For photons the ultra-relativistic case always holds, so that expression (1.5) is valid. This means that it makes sense to speak of the coordinates of a photon only in those cases where the characteristic dimensions are large compared with the wavelength. But this is nothing other than the "classical" limiting case, corresponding to geometrical optics, in which one may speak of the propagation of light along definite trajectories — rays. In the quantum case, however, when the wavelength cannot be regarded as small, the concept of the coordinates of a photon becomes vacuous. We shall see later (see § 4) that in the mathematical formalism of the theory the non-measurability of the photon's coordinates already shows itself in the impossibility of forming from its wave function a quantity that could play the part of a probability density satisfying the requirements which relativistic invariance demands.

On the basis of all that has been said, it is natural to suppose that the future theory will renounce altogether the consideration of the time development of the processes of interaction of particles. It will show that in these processes no exactly definable characteristics exist (even within the limits of the usual quantum-mechanical accuracy), so that the description of a process in time will turn out to be just as illusory as classical trajectories proved to be in non-relativistic quantum mechanics. The only observable quantities will be the characteristics (momenta, polarizations) of free particles — the initial particles that enter into the interaction and the final particles that have arisen as a result of the process (*L. D. Landau, R. Peierls*, 1930).

The characteristic formulation of a problem in relativistic quantum theory consists in the determination of the probability amplitudes of transitions connecting given initial and final states (that is, at $t \to \mp\infty$) of a system of particles. The totality of the transition amplitudes between all possible states constitutes the *scattering matrix*, or *S-matrix*. This matrix will be the carrier of all the information about the processes of interaction of particles that possesses an observable physical meaning (*W. Heisenberg*, 1938).

At the present time a complete, logically closed relativistic quantum theory does not yet exist. We shall see that the existing theory introduces new physical aspects into the manner of describing the states of particles, which acquires certain features of field theory (see § 10). The theory is constructed, however, to a considerable degree on the model of, and with the help of the concepts of, ordinary quantum mechanics. This way of building the theory has led to success in the domain of quantum electrodynamics. The absence of complete logical closure shows itself in this theory in the existence of divergent expressions when its mathematical apparatus is applied directly; but for the removal of these divergences there exist entirely unambiguous procedures. Nevertheless these procedures retain to a considerable degree the character of semi-empirical recipes, and our confidence in the correctness of the results obtained in this way rests, in the end, on their excellent agreement with experiment, and not on the internal consistency and logical coherence of the fundamental principles of the theory.

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the electron; we thus set aside coordinate measurements by collision, in which over the time of observation the result occurs with a probability short of unity. In such a case a deflection of the particle does allow the electron's position to be inferred, but from the absence of a deflection nothing at all can be concluded.

CHAPTER I

Photon

§ 2. Quantization of the Free Electromagnetic Field

In order to treat the electromagnetic field as a quantum object, it is convenient to start from a classical description in which the field is characterized by a discrete, albeit infinite, set of variables; such a description will allow one to apply the standard formalism of quantum mechanics directly. A description by means of potentials specified at every point of space is, in essence, a description in terms of a continuous set of variables.

Let $\mathbf{A}(\mathbf{r}, t)$ be the vector potential of the free electromagnetic field, satisfying the "transversality condition"

$$\nabla \cdot \mathbf{A} = 0. \tag{2.1}$$

The scalar potential then vanishes, $\Phi = 0$, and the fields $\mathbf{E}$ and $\mathbf{H}$ are

$$\mathbf{E} = -\dot{\mathbf{A}}, \qquad \mathbf{H} = \nabla \times \mathbf{A}. \tag{2.2}$$

Maxwell's equations reduce to the wave equation for $\mathbf{A}$:

$$\Delta \mathbf{A} - \frac{\partial^2 \mathbf{A}}{\partial t^2} = 0. \tag{2.3}$$

As is well known (see Vol. II, § 52), in classical electrodynamics the passage to a description in terms of a discrete set of variables is effected by considering the field inside a large but finite volume V.[1] Let us recall, omitting the details of the calculation, how this is done.

A field in a finite volume can be expanded in travelling plane waves, so that its potential takes the form of a series

$$\mathbf{A} = \sum_{\mathbf{k}} (\mathbf{a}_{\mathbf{k}} e^{i\mathbf{kr}} + \mathbf{a}_{\mathbf{k}}^* e^{-i\mathbf{kr}}), \tag{2.4}$$

where the coefficients $\mathbf{a}_{\mathbf{k}}$ depend on time according to

$$\mathbf{a}_{\mathbf{k}} \sim e^{-i\omega t}, \quad \omega = |\mathbf{k}|. \tag{2.5}$$

By virtue of condition (2.1) the complex vectors $\mathbf{a}_{\mathbf{k}}$ are orthogonal to the corresponding wave vectors: $\mathbf{a}_{\mathbf{k}}\mathbf{k} = 0$.

The summation in (2.4) runs over an infinite discrete set of values of the wave vector (its three components k_x, k_y, k_z). The transition to an integration over a continuous distribution is made with the aid of the expression

$$\frac{d^3 k}{(2\pi)^3}$$

[1] To avoid burdening formulae with superfluous factors, we set $V = 1$.

for the number of allowed values of $\mathbf{k}$ falling within a volume element $d^3k = dk_x\, dk_y\, dk_z$ of $\mathbf{k}$-space.

Once the vectors $\mathbf{a_k}$ are given, the field inside the volume in question is completely determined. One may therefore view these quantities as a discrete set of classical "field variables". In order to find the passage to a quantum theory, however, a further transformation of these quantities is needed, one that brings the field equations into a form analogous to the canonical (Hamiltonian) equations of classical mechanics. The canonical variables of the field are introduced by means of

$$\mathbf{Q_k} = \frac{1}{\sqrt{4\pi}}(\mathbf{a_k} + \mathbf{a_k^*}), \qquad \mathbf{P_k} = -\frac{i\omega}{\sqrt{4\pi}}(\mathbf{a_k} - \mathbf{a_k^*}) = \dot{\mathbf{Q}}_\mathbf{k} \tag{2.6}$$

(they are evidently real). The vector potential is expressed through the canonical variables as

$$\mathbf{A} = \sqrt{4\pi}\sum_\mathbf{k}\left(\mathbf{Q_k}\cos\mathbf{kr} - \frac{1}{\omega}\mathbf{P_k}\sin\mathbf{kr}\right). \tag{2.7}$$

To find the Hamiltonian H one must evaluate the total field energy

$$\frac{1}{8\pi}\int(\mathbf{E}^2 + \mathbf{H}^2)\, d^3x,$$

expressed in terms of the quantities $\mathbf{Q_k}$, $\mathbf{P_k}$. Substituting the expansion (2.7) for $\mathbf{A}$, computing $\mathbf{E}$ and $\mathbf{H}$ by means of (2.2), and integrating, one obtains

$$H = \frac{1}{2}\sum_\mathbf{k}(\mathbf{P_k^2} + \omega^2\mathbf{Q_k^2}).$$

Each of the vectors $\mathbf{P_k}$ and $\mathbf{Q_k}$ is perpendicular to the wave vector $\mathbf{k}$ and therefore has two independent components. The direction of these vectors fixes the polarization direction of the corresponding wave. Denoting the two components of $\mathbf{Q_k}$, $\mathbf{P_k}$ (in the plane perpendicular to $\mathbf{k}$) by $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$ ($\alpha = 1, 2$), we rewrite the Hamiltonian as

$$H = \sum_{\mathbf{k}\alpha}\frac{1}{2}(P_{\mathbf{k}\alpha}^2 + \omega^2 Q_{\mathbf{k}\alpha}^2). \tag{2.8}$$

The Hamiltonian has thus broken up into a sum of independent terms, each involving just one pair $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$. Every term answers to a travelling wave of a given wave vector and polarization and reproduces the structure of a single harmonic-oscillator Hamiltonian. One accordingly speaks of this decomposition as the expansion of the field in *oscillators*.

We now turn to the quantization of the free electromagnetic field. The method of classical description just set out makes the route to quantum theory obvious. The canonical variables — the generalized coordinates $Q_{\mathbf{k}\alpha}$ and generalized momenta $P_{\mathbf{k}\alpha}$ — are now to be treated as operators obeying the commutation rule

$$\hat{P}_{\mathbf{k}\alpha}\hat{Q}_{\mathbf{k}\alpha} - \hat{Q}_{\mathbf{k}\alpha}\hat{P}_{\mathbf{k}\alpha} = -i \tag{2.9}$$

(operators with different $\mathbf{k}\alpha$ all commute with one another). Together with them the potential $\mathbf{A}$ and, by (2.2), the field strengths $\mathbf{E}$ and $\mathbf{H}$ also become (Hermitian) operators.

A consistent determination of the field Hamiltonian requires evaluation of the integral

$$\hat{H} = \frac{1}{8\pi} \int (\hat{\mathbf{E}}^2 + \hat{\mathbf{H}}^2)\, d^3x, \tag{2.10}$$

in which $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ are expressed through the operators $\hat{P}_{\mathbf{k}\alpha}$, $\hat{Q}_{\mathbf{k}\alpha}$. In practice, however, the non-commutativity of the latter does not make itself felt here, since the products $Q_{\mathbf{k}\alpha} P_{\mathbf{k}\alpha}$ enter with the factor $\cos \mathbf{kr} \cdot \sin \mathbf{kr}$, which vanishes upon integration over the full volume. Consequently the Hamiltonian takes the form

$$\hat{H} = \sum_{\mathbf{k}\alpha} \frac{1}{2} (\hat{P}^2_{\mathbf{k}\alpha} + \omega^2 \hat{Q}^2_{\mathbf{k}\alpha}), \tag{2.11}$$

exactly reproducing the classical Hamiltonian, as one would naturally expect.

Finding the eigenvalues of this Hamiltonian requires no special computation, since the problem reduces to the familiar one of the energy levels of linear oscillators (see Vol. III, § 23). We can therefore write down at once the energy levels of the field:

$$E = \sum_{\mathbf{k}\alpha} \left(N_{\mathbf{k}\alpha} + \frac{1}{2} \right) \omega, \tag{2.12}$$

where $N_{\mathbf{k}\alpha}$ are non-negative integers.

Discussion of this formula will be taken up in the next section; here we write out the matrix elements of the quantities $Q_{\mathbf{k}\alpha}$, which can be obtained directly from the known formulae for the matrix elements of the oscillator coordinate (see Vol. III, § 23). The non-vanishing matrix elements are

$$\langle N_{\mathbf{k}\alpha} | Q_{\mathbf{k}\alpha} | N_{\mathbf{k}\alpha} - 1 \rangle = \sqrt{\frac{N_{\mathbf{k}\alpha}}{2\omega}}. \tag{2.13}$$

The matrix elements of $P_{\mathbf{k}\alpha} = \dot{Q}_{\mathbf{k}\alpha}$ differ from those of $Q_{\mathbf{k}\alpha}$ only by a factor $\pm i\omega$.

For subsequent calculations it will, however, be more convenient to work not with $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$ themselves but with their linear combinations $\omega Q_{\mathbf{k}\alpha} \pm i P_{\mathbf{k}\alpha}$, which have matrix elements only for the transitions $N_{\mathbf{k}\alpha} \to N_{\mathbf{k}\alpha} \pm 1$. Accordingly we introduce the operators

$$\hat{c}_{\mathbf{k}\alpha} = \frac{1}{\sqrt{2\omega}} (\omega \hat{Q}_{\mathbf{k}\alpha} + i \hat{P}_{\mathbf{k}\alpha}), \qquad \hat{c}^+_{\mathbf{k}\alpha} = \frac{1}{\sqrt{2\omega}} (\omega \hat{Q}_{\mathbf{k}\alpha} - i \hat{P}_{\mathbf{k}\alpha}) \tag{2.14}$$

(the classical quantities $c_{\mathbf{k}\alpha}$, $c^*_{\mathbf{k}\alpha}$ coincide, up to a factor $\sqrt{2\pi/\omega}$, with the coefficients $a_{\mathbf{k}\alpha}$, $a^*_{\mathbf{k}\alpha}$ in the expansion (2.4)).

The matrix elements of these operators are

$$\langle N_{\mathbf{k}\alpha} - 1 | \hat{c}_{\mathbf{k}\alpha} | N_{\mathbf{k}\alpha} \rangle = \langle N_{\mathbf{k}\alpha} | \hat{c}^+_{\mathbf{k}\alpha} | N_{\mathbf{k}\alpha} - 1 \rangle = \sqrt{N_{\mathbf{k}\alpha}}. \tag{2.15}$$

The commutation rule between $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}^+_{\mathbf{k}\alpha}$ follows from the definition (2.14) and the rule (2.9):

$$\hat{c}_{\mathbf{k}\alpha} \hat{c}^+_{\mathbf{k}\alpha} - \hat{c}^+_{\mathbf{k}\alpha} \hat{c}_{\mathbf{k}\alpha} = 1. \tag{2.16}$$

Returning to the vector potential, we adopt an expansion of the same type as (2.4), but with the coefficients now promoted to operators:

$$\hat{\mathbf{A}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} \mathbf{A}_{\mathbf{k}\alpha} + \hat{c}^+_{\mathbf{k}\alpha} \mathbf{A}^*_{\mathbf{k}\alpha}), \tag{2.17}$$

where

$$\mathbf{A}_{\mathbf{k}\alpha} = \sqrt{4\pi} \, \frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}} \, e^{i\mathbf{kr}}. \tag{2.18}$$

Here $\mathbf{e}^{(\alpha)}$ denotes a unit vector specifying the polarization direction of the oscillator; the vectors $\mathbf{e}^{(\alpha)}$ are perpendicular to the wave vector $\mathbf{k}$, and for each $\mathbf{k}$ there are two independent polarizations.

Analogously, for the operators $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ we write

$$\hat{\mathbf{E}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} \mathbf{E}_{\mathbf{k}\alpha} + \hat{c}^+_{\mathbf{k}\alpha} \mathbf{E}^*_{\mathbf{k}\alpha}), \quad \hat{\mathbf{H}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} \mathbf{H}_{\mathbf{k}\alpha} + \hat{c}^+_{\mathbf{k}\alpha} \mathbf{H}^*_{\mathbf{k}\alpha}), \tag{2.19}$$

with

$$\mathbf{E}_{\mathbf{k}\alpha} = i\omega \mathbf{A}_{\mathbf{k}\alpha}, \qquad \mathbf{H}_{\mathbf{k}\alpha} = \mathbf{n} \times \mathbf{E}_{\mathbf{k}\alpha} \qquad (\mathbf{n} = \mathbf{k}/\omega). \tag{2.20}$$

The vectors $\mathbf{A}_{\mathbf{k}\alpha}$ are mutually orthogonal in the sense that

$$\int \mathbf{A}_{\mathbf{k}\alpha} \mathbf{A}^*_{\mathbf{k}'\alpha'} \, d^3x = \frac{2\pi}{\omega} \, \delta_{\alpha\alpha'} \, \delta_{\mathbf{k}\mathbf{k}'}. \tag{2.21}$$

To see this, note that when $\mathbf{A}_{\mathbf{k}\alpha}$ and $\mathbf{A}^*_{\mathbf{k}'\alpha'}$ carry different wave vectors their product involves $e^{i(\mathbf{k}-\mathbf{k}')\mathbf{r}}$, which integrates to zero over the volume; when only the polarizations differ, the orthogonality $\mathbf{e}^{(\alpha)}\mathbf{e}^{(\alpha')*} = 0$ of the two independent polarization directions provides the required vanishing. Analogous relations hold for the vectors $\mathbf{E}_{\mathbf{k}\alpha}$, $\mathbf{H}_{\mathbf{k}\alpha}$. Their normalization is conveniently written as

$$\frac{1}{4\pi} \int (\mathbf{E}_{\mathbf{k}\alpha} \mathbf{E}^*_{\mathbf{k}'\alpha'} + \mathbf{H}_{\mathbf{k}\alpha} \mathbf{H}^*_{\mathbf{k}'\alpha'}) \, d^3x = \omega \, \delta_{\mathbf{k}\mathbf{k}'} \, \delta_{\alpha\alpha'}. \tag{2.22}$$

Substituting the operators (2.19) into (2.10) and integrating with the help of (2.22), we get the field Hamiltonian expressed through the operators $\hat{c}$, $\hat{c}^+$:

$$\hat{H} = \sum_{\mathbf{k}\alpha} \frac{1}{2} \omega (\hat{c}_{\mathbf{k}\alpha} \hat{c}^+_{\mathbf{k}\alpha} + \hat{c}^+_{\mathbf{k}\alpha} \hat{c}_{\mathbf{k}\alpha}). \tag{2.23}$$

In the representation under discussion (with the matrix elements of $\hat{c}$, $\hat{c}^+$ given by (2.15)) this operator is diagonal, and its eigenvalues coincide, naturally, with (2.12).

In classical theory the field momentum is defined by the integral

$$\mathbf{P} = \frac{1}{4\pi} \int \mathbf{E} \times \mathbf{H} \, d^3x.$$

On passing to the quantum theory by replacing $\mathbf{E}$ and $\mathbf{H}$ with the operators (2.19), one readily finds

$$\hat{\mathbf{P}} = \sum_{\mathbf{k}\alpha} \frac{1}{2} (\hat{P}^2_{\mathbf{k}\alpha} + \omega^2 \hat{Q}^2_{\mathbf{k}\alpha}) \mathbf{n} \tag{2.24}$$

— which mirrors the familiar classical link between energy and momentum of plane waves. This operator has eigenvalues

$$\mathbf{P} = \sum_{\mathbf{k}\alpha} \mathbf{k}\left(N_{\mathbf{k}\alpha} + \frac{1}{2}\right). \tag{2.25}$$

The operator representation realized by the matrix elements (2.15) is called the "occupation-number representation": it amounts to characterizing the state of the system (the field) through the quantum numbers $N_{\mathbf{k}\alpha}$ (*occupation numbers*). Within this representation the field operators (2.19) (and hence the Hamiltonian (2.11)) act on the system's wave function, written as a function of the numbers $N_{\mathbf{k}\alpha}$; we denote it $\Phi(N_{\mathbf{k}\alpha}, t)$. The field operators (2.19) do not depend on time explicitly. This corresponds to the Schrödinger representation of operators, familiar from non-relativistic quantum mechanics. What does depend on time is the state of the system, $\Phi(N_{\mathbf{k}\alpha}, t)$, its time dependence being governed by the Schrödinger equation

$$i\frac{\partial \Phi}{\partial t} = \hat{H}\Phi.$$

This description of the field is relativistically invariant in essence, since it rests on the invariant Maxwell equations. Yet the invariance is not exhibited explicitly — above all because the spatial coordinates and the time enter the description in a highly asymmetric way.

Within a relativistic framework it is desirable to give the description a more overtly invariant appearance. For this purpose one uses the so-called Heisenberg representation, wherein the explicit dependence on time is shifted onto the operators (see Vol. III, § 13). Time and spatial coordinates then appear symmetrically in the field-operator expressions, while the system state Φ depends only on the occupation numbers.

For the operator $\hat{\mathbf{A}}$ the passage to the Heisenberg representation amounts to replacing, in each term of the sum in (2.17), (2.18), the factor $e^{i\mathbf{kr}}$ by $e^{i(\mathbf{kr}-\omega t)}$; that is, one is to understand by $\mathbf{A}_{\mathbf{k}\alpha}$ the time-dependent functions

$$\mathbf{A}_{\mathbf{k}\alpha} = \sqrt{4\pi}\,\frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}}\,e^{-i(\omega t-\mathbf{kr})}. \tag{2.26}$$

This is easily verified by noting that the matrix element of a Heisenberg operator for the transition $i \to f$ must contain the factor $\exp\{-i(E_i - E_f)t\}$, where E_i and E_f are the energies of the initial and final states (see Vol. III, § 13). For a transition in which $N_{\mathbf{k}}$ decreases or increases by one, this factor reduces to $e^{-i\omega t}$ or $e^{i\omega t}$, respectively. The indicated replacement satisfies this requirement.

Henceforth (in treating both the electromagnetic field and the fields of particles) we shall always understand the Heisenberg representation of operators.

§ 3. Photons

Let us now turn to a discussion of the quantization formulae derived above.

To begin with, formula (2.12) for the energy of the field exposes the following difficulty. At the lowest energy level, all oscillator quantum numbers $N_{\mathbf{k}\alpha}$ vanish (this state is known as the *electromagnetic field vacuum*). Yet even in this state every oscillator retains a nonzero "zero-point energy" $\omega/2$. Summing over the entire infinite set of oscillators then produces an infinite result. One is thus confronted with one of the "divergences" stemming from the absence of full logical self-consistency in the existing theory.

As long as one is concerned only with the eigenvalues of the field energy, this difficulty can be eliminated by simply subtracting away the zero-point oscillation energy, i.e. by writing for the energy and momentum of the field[2]

$$E = \sum_{\mathbf{k}\alpha} N_{\mathbf{k}\alpha}\omega, \qquad \mathbf{P} = \sum_{\mathbf{k}\alpha} N_{\mathbf{k}\alpha}\mathbf{k}. \tag{3.1}$$

These formulae make it possible to introduce the concept fundamental to all of quantum electrodynamics: that of *light quanta*, or *photons*[3]. Namely, the free electromagnetic field can be regarded as a collection of particles, each carrying energy $\omega\ (= \hbar\omega)$ and momentum $\mathbf{k}\ (= \mathbf{n}\hbar\omega/c)$. The relationship between a photon's energy and momentum is precisely what relativistic mechanics requires for particles of zero rest mass that propagate at the speed of light. The occupation numbers $N_{\mathbf{k}\alpha}$ acquire the meaning of the number of photons with given momenta $\mathbf{k}$ and polarizations $\mathbf{e}^{(\alpha)}$. The polarization property of a photon is analogous to the notion of spin for other particles (the specific features of the photon in this regard will be examined below, in § 6).

It is easy to see that the mathematical formalism developed in the preceding section is fully consistent with the picture of the electromagnetic field as an assembly of photons; it is nothing other than the apparatus of so-called second quantization as applied to a system of photons[4]. In this method (see Vol. III, § 64) the occupation numbers serve as the independent variables, and the operators act on functions of those numbers. The central role is played by the "annihilation" and "creation" operators for particles, which respectively decrease or increase an occupation number by one. Exactly such operators are $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}^+_{\mathbf{k}\alpha}$: the operator $\hat{c}_{\mathbf{k}\alpha}$ destroys a photon in the state $\mathbf{k}\alpha$, while $\hat{c}^+_{\mathbf{k}\alpha}$ creates one in that state.

The commutation rule (2.16) corresponds to particles obeying Bose statistics. Photons are therefore bosons, as one might have anticipated: the number of photons occupying any given state can be arbitrary (we shall return to the significance of this fact in § 5).

The plane waves $\mathbf{A}_{\mathbf{k}\alpha}$ (2.26), which appear in the operator $\hat{\mathbf{A}}$ (2.17) as coefficients multiplying the photon annihilation operators, can be interpreted as the wave functions of photons possessing definite momenta $\mathbf{k}$ and polarizations $\mathbf{e}^{(\alpha)}$. This interpretation corresponds to expanding the ψ-operator as a series in the stationary-state wave functions of the particle within the non-relativistic second-quantization framework (unlike the

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[2]This subtraction can be carried out in a formally consistent manner by agreeing to interpret the operator products in (2.10) as "normal-ordered", i.e. arranged so that the operators $\hat{c}^+$ always stand to the left of the operators $\hat{c}$. Formula (2.23) then takes the form

$$\hat{H} = \sum_{\mathbf{k}\alpha}\big(\omega\,\hat{c}^+_{\mathbf{k}\alpha}\hat{c}_{\mathbf{k}\alpha}\big).$$

[3]The photon concept was first put forward by *Einstein* (A. Einstein, 1905).

[4]The method of second quantization as applied to radiation theory was developed by *Dirac* (P. A. M. Dirac, 1927).

non-relativistic case, however, the expansion (2.17) involves both annihilation and creation operators; the significance of this distinction will become clear later, see § 12).

The wave function (2.26) is normalized by the condition

$$\int \frac{1}{4\pi}\left(|\mathbf{E}_{\mathbf{k}\alpha}|^2 + |\mathbf{H}_{\mathbf{k}\alpha}|^2\right) d^3x = \omega \tag{3.2}$$

This amounts to normalizing to one photon in a volume $V = 1$. Indeed, the integral on the left-hand side of the equation represents the quantum-mechanical expectation value of the photon's energy in a state described by the given wave function[5]. On the right-hand side of (3.2) stands the energy of a single photon.

The role of the "Schrödinger equation" for the photon is played by Maxwell's equations. In the present case (for the potential $\mathbf{A}(\mathbf{r}, t)$ satisfying condition (2.1)) this is the wave equation

$$\frac{\partial^2 \mathbf{A}}{\partial t^2} - \Delta \mathbf{A} = 0.$$

In the general case of arbitrary stationary states, the photon "wave functions" are complex solutions of this equation whose time dependence enters through the factor $e^{-i\omega t}$.

Speaking of the photon wave function, let us emphasize once more that it cannot by any means be treated as a probability amplitude for the spatial localization of a photon — in contrast to the basic meaning of the wave function in non-relativistic quantum mechanics. This is linked to the fact that (as noted in § 1) the notion of photon coordinates has no physical meaning at all. We shall revisit the mathematical side of this issue at the end of the next section.

The Fourier-transform components of $\mathbf{A}(\mathbf{r}, t)$ with respect to the coordinates constitute the photon wave function in the momentum representation; we denote it $\mathbf{A}(\mathbf{k}, t) = \mathbf{A}(\mathbf{k})e^{-i\omega t}$. Thus, for a state with a definite momentum $\mathbf{k}$ and polarization $\mathbf{e}^{(\alpha)}$ the momentum-representation wave function is given simply by the coefficient of the exponential factor in (2.26):

$$\mathbf{A}_{\mathbf{k}\alpha}(\mathbf{k}', \alpha') = \sqrt{4\pi}\,\frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}}\,\delta_{\mathbf{k}'\mathbf{k}}\delta_{\alpha\alpha'}. \tag{3.3}$$

In keeping with the measurability of the momentum of a free particle, the momentum-representation wave function carries deeper physical content than its coordinate-space counterpart: it provides a way to compute the probabilities $w_{\mathbf{k}\alpha}$ of the various values of momentum and polarization for a photon in a given state. According to the general rules of quantum mechanics, $w_{\mathbf{k}\alpha}$ is given by the squared modulus of the expansion coefficients of $\mathbf{A}(\mathbf{k}')$ in the wave functions of states with definite $\mathbf{k}$ and $\mathbf{e}^{(\alpha)}$:

$$w_{\mathbf{k}\alpha} \propto \left|\sum_{\mathbf{k}'\alpha'} \mathbf{A}^*_{\mathbf{k}\alpha}(\mathbf{k}', \alpha')\mathbf{A}(\mathbf{k}')\right|^2$$

[5]Note that the coefficient $1/(4\pi)$ in the integral (3.2) is twice the usual coefficient $1/(8\pi)$ in (2.10). This difference is ultimately connected with the complex character of the vectors $\mathbf{E}_{\mathbf{k}\alpha}$, $\mathbf{H}_{\mathbf{k}\alpha}$, as opposed to the Hermitian field operators $\hat{\mathbf{E}}$, $\hat{\mathbf{H}}$.

(the proportionality coefficient depends on the choice of normalization for the functions). Substituting (3.3), we obtain

$$w_{\mathbf{k}\alpha} \propto |\mathbf{e}^{(\alpha)}\mathbf{A}(\mathbf{k})|^2. \tag{3.4}$$

Summing over the two polarizations, we find the probability that a photon has momentum **k**:

$$w_{\mathbf{k}} \propto |\mathbf{A}(\mathbf{k})|^2. \tag{3.5}$$

§ 4. Gauge invariance

As is well known, the choice of field potentials in classical electrodynamics is not unique: the components of the 4-potential A_μ may be subjected to an arbitrary *gauge* (or *gradient*) transformation of the form

$$A_\mu \to A_\mu + \partial_\mu \chi, \tag{4.1}$$

where χ is an arbitrary function of the coordinates and time (see Vol. II, § 18). In the case of a plane wave, restricting oneself to transformations that preserve the functional form of the potential (its proportionality to $\exp\left(-ik_\mu x^\mu\right)$), the non-uniqueness amounts to the freedom of adding to the wave amplitude an arbitrary 4-vector proportional to k^μ.

The ambiguity of the potential persists, of course, in the quantum theory as well — as applied to the field operators or to the photon wave functions. Without prejudging the choice of potentials, one should write in place of (2.17) an analogous expansion for the operator 4-potential

$$\hat{A}^\mu = \sum_{\mathbf{k}\alpha}\left(\hat{c}_{\mathbf{k}\alpha}A^\mu_{\mathbf{k}\alpha} + \hat{c}^+_{\mathbf{k}\alpha}A^{\mu*}_{\mathbf{k}\alpha}\right), \tag{4.2}$$

where the wave functions $A^\mu_{\mathbf{k}\alpha}$ are 4-vectors of the form

$$A^\mu_{\mathbf{k}} = \sqrt{4\pi}\,\frac{e^\mu}{\sqrt{2\omega}}\,e^{-ik_\nu x^\nu}, \qquad e_\mu e^{\mu*} = -1,$$

or, in a compact notation that suppresses the four-dimensional vector indices:

$$A_{\mathbf{k}} = \sqrt{4\pi}\,\frac{e}{\sqrt{2\omega}}\,e^{-ikx}, \qquad ee^* = -1. \tag{4.3}$$

Here the 4-momentum is $k_\mu = (\omega, \mathbf{k})$ (so that $kx = \omega t - \mathbf{kr}$), and e is the unit polarization 4-vector[6].

If one restricts oneself to gauge transformations that do not alter how the function (4.3) depends on the coordinates and time, they amount to the replacement

$$e_\mu \to e_\mu + \chi k_\mu, \tag{4.4}$$

[6]Expression (4.3) does not have an entirely relativistically covariant (4-vector) form, which is related to the non-invariant character of our normalization to a finite volume $V = 1$. This is, however, of no fundamental significance and is fully compensated by the convenience of such a normalization scheme. We shall see later that it ensures simple and automatic derivation of real physical quantities in the proper invariant form.

where $\chi = \chi(k^\mu)$ is an arbitrary function. The transversality of the polarization means that one can always choose a gauge in which the 4-vector e takes the form

$$e^\mu = (0, \mathbf{e}), \qquad \mathbf{ek} = 0 \tag{4.5}$$

(we shall call such a gauge *three-dimensionally transverse*). In an invariant four-dimensional notation this requirement is written as the condition of *four-dimensional transversality*

$$ek = 0. \tag{4.6}$$

Note that this condition (as well as the normalization condition $ee^* = -1$) is not violated by the transformation (4.4), by virtue of $k^2 = 0$. On the other hand, the vanishing of the 4-momentum squared of a particle signifies that its mass is zero. A direct link between gauge invariance and the vanishing of the photon mass is thereby revealed (further aspects of this connection will be discussed in § 14).

No measurable physical quantity should change under a gauge transformation of the wave functions of the photons taking part in a given process. This requirement of *gauge invariance* plays an even greater role in quantum electrodynamics than in the classical theory. As we shall see from numerous examples, it serves here, alongside the requirement of relativistic invariance, as a powerful heuristic principle.

In turn, the gauge invariance of the theory is closely bound up with the law of conservation of electric charge; we shall dwell on this aspect in § 43.

We already remarked in the preceding section that the coordinate-space wave function of a photon cannot be interpreted as a probability amplitude for its spatial localization. From a mathematical standpoint this circumstance manifests itself in the impossibility of constructing, out of the wave function, a quantity whose formal properties alone would qualify it to serve as a probability density. Such a quantity would have to be an essentially positive bilinear combination of the wave function A_μ and its complex conjugate. Moreover, it would need to satisfy definite requirements of relativistic covariance — namely, to be the time component of a 4-vector (the point being that the continuity equation expressing conservation of particle number takes the four-dimensional form of a vanishing 4-divergence of the current 4-vector; the time component of this current is, in the present context, the probability density for the localization of the particle; see Vol. II, § 29). On the other hand, gauge invariance demands that A_μ enter the current only through the antisymmetric tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = -i(k_\mu A_\nu - k_\nu A_\mu)$. The current 4-vector would therefore have to be assembled bilinearly from $F_{\mu\nu}$ and $F^*_{\mu\nu}$ (together with components of the 4-vector k_μ). But no such 4-vector can be constructed at all: every expression satisfying the stated conditions (for instance, $k^\lambda F^*_{\mu\nu} F_\lambda{}^\nu$) vanishes by the transversality condition ($k^\lambda F_{\nu\lambda} = 0$), quite apart from the fact that it would not be essentially positive (since it contains odd powers of the components k_μ).

§ 5. The electromagnetic field in quantum theory

Treating the field as a collection of photons is the sole description that fully captures the physical meaning of the electromagnetic field within quantum theory. It replaces the classical characterization through field strengths. The field strengths themselves enter the mathematical apparatus of the photon picture as operators of second quantization.

It is well known that a quantum system behaves classically when the quantum numbers characterizing its stationary states become large. For the free electromagnetic field (within a given volume) this requires that the oscillator quantum numbers — that is, the photon occupation numbers $N_{\mathbf{k}\alpha}$ — be large. The fact that photons obey Bose statistics is of profound importance in this context. In the mathematical formalism, the link between Bose statistics and classical field behavior is encoded in the commutation rules for $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}^+_{\mathbf{k}\alpha}$. When $N_{\mathbf{k}\alpha}$ is large, so that the matrix elements of these operators are also large, the unity appearing on the right-hand side of the commutation relation (2.16) may be dropped, giving

$$\hat{c}_{\mathbf{k}\alpha}\hat{c}^+_{\mathbf{k}\alpha} \simeq \hat{c}^+_{\mathbf{k}\alpha}\hat{c}_{\mathbf{k}\alpha},$$

i.e. the operators go over into mutually commuting classical quantities $c_{\mathbf{k}\alpha}$, $c^*_{\mathbf{k}\alpha}$ that determine the classical field strengths.

The quasi-classicality condition for the field requires, however, a further refinement. The point is that if all the numbers $N_{\mathbf{k}\alpha}$ are large, then upon summing over every state $\mathbf{k}\alpha$ the field energy is in any case infinite, rendering the condition meaningless.

A physically sensible formulation of the quasi-classicality criterion rests on examining values of the field averaged over some small time interval Δt. If one writes the classical electric field $\bar{\mathbf{E}}$ (or the magnetic field $\bar{\mathbf{H}}$) as a Fourier integral over time, then upon time-averaging over an interval Δt only those Fourier components whose frequencies satisfy $\omega\Delta t \lesssim 1$ contribute appreciably to the mean value of $\bar{\mathbf{E}}$; in the opposite limit the oscillating factor $e^{-i\omega t}$ nearly averages to zero. Consequently, in establishing the quasi-classicality condition for the time-averaged field, one need consider only those quantum oscillators whose frequencies satisfy $\omega \lesssim 1/\Delta t$. It suffices to require that the quantum numbers of these oscillators be large. The number of oscillators with frequencies between zero and $\omega \sim 1/\Delta t$ (referred to the volume $V = 1$) is of order[7]

$$\left(\frac{\omega}{c}\right)^3 \sim \frac{1}{(c\Delta t)^3}. \tag{5.1}$$

The total energy of the field per unit volume is of order $\bar{\mathbf{E}}^2$. Dividing this quantity by the number of oscillators and by a characteristic energy of one photon ($\sim \hbar\omega$), one obtains an estimate for the photon occupation numbers

$$N_{\mathbf{k}} \sim \frac{\bar{\mathbf{E}}^2 c^3}{\hbar\omega^4}.$$

Demanding that this number be large, we arrive at the inequality

$$|\mathbf{E}| \gg \frac{\sqrt{\hbar c}}{(c\Delta t)^2}. \tag{5.2}$$

This is the sought-after condition permitting a classical treatment of the time-averaged (over intervals Δt) field. We see that the field must be sufficiently strong — all the stronger, the shorter the averaging interval Δt. For time-varying fields this interval must not, of course, exceed the characteristic periods over which the field changes appreciably.

[7] In this section we use ordinary units.

Consequently, sufficiently weak time-varying fields can under no circumstances be quasi-classical. Only for static (time-independent) fields may one set $\Delta t \to \infty$, so that the right-hand side of the inequality (5.2) vanishes. This means that a static field is always classical.

It has already been noted that the classical expressions for the electromagnetic field in the form of a superposition of plane waves must be treated as operator-valued in the quantum theory. The physical meaning of these operators is, however, rather limited. Indeed, a physically meaningful field operator ought to yield vanishing field values in the photon-vacuum state. Yet the expectation value of the squared-field operator $\hat{\mathbf{E}}^2$ in the ground state, which coincides up to a factor with the zero-point energy of the field, turns out to be infinite (by "expectation value" we mean the quantum-mechanical average, i.e. the corresponding diagonal matrix element of the operator). This divergence cannot be circumvented even by any formal subtraction procedure (as can be done for the field energy), since in the present case we would need to carry out the subtraction through some sensible modification of the operators $\hat{\mathbf{E}}$, $\hat{\mathbf{H}}$ themselves (rather than their squares), which proves to be impossible.

§ 6. Angular momentum and parity of the photon

As with any particle, the photon can possess a definite angular momentum. In order to elucidate the properties of this quantity for the photon, let us begin by recalling how the angular momentum of a particle is connected with its wave function in the formalism of quantum mechanics.

The total angular momentum $\mathbf{j}$ of a particle consists of the orbital part $\mathbf{l}$ and the intrinsic part — the spin $\mathbf{s}$. A particle of spin s is described by a symmetric spinor of rank $2s$, i.e. a set of $2s + 1$ components that transform into one another under coordinate-system rotations according to a specific law. The orbital angular momentum is instead linked to how the wave functions depend on coordinates: states with orbital angular momentum l have wave-function components that are expressed (linearly) in terms of spherical harmonics of order l.

The ability to distinguish consistently between spin and orbital angular momentum therefore demands that the "spin" and "coordinate" properties of the wave functions be independent: the coordinate dependence of each spinor component (at a given instant of time) should not be constrained by any supplementary conditions. Passing to the momentum representation, coordinate dependence becomes dependence on $\mathbf{k}$. In the three-dimensionally transverse gauge the photon wave function is the vector $\mathbf{A}(\mathbf{k})$. Because a vector is equivalent to a rank-2 spinor, one could formally ascribe spin 1 to the photon. This vector wave function, however, must satisfy the transversality requirement $\mathbf{k}\mathbf{A}(\mathbf{k}) = 0$, an extra constraint on the way $\mathbf{A}(\mathbf{k})$ depends on momentum. It follows that the dependence on $\mathbf{k}$ can no longer be chosen freely for every vector component at once, making it impossible to disentangle orbital angular momentum and spin.

We note also that for the photon one cannot define spin as the angular momentum of the particle at rest, since for a photon traveling at the speed of light no rest frame exists at all.

Thus, for a photon one can speak only of its total angular momentum. It is clear from the outset that the total angular momentum can assume only integer values. This is already evident from the fact that among the quantities characterizing the photon there are no spinors of odd rank.

As for any particle, the state of a photon is further characterized by its parity, associated with the behavior of the wave function under spatial inversion (see Vol. III, § 30). In the momentum representation, a reversal of the sign of the coordinates corresponds to reversing the sign of every component of $\mathbf{k}$. The action of the inversion operator $\hat{P}$ on a scalar function $\varphi(\mathbf{k})$ amounts simply to this sign change: $\hat{P}\varphi(\mathbf{k}) = \varphi(-\mathbf{k})$. When acting on the vector function $\mathbf{A}(\mathbf{k})$, one must additionally take into account that reversing the axes also reverses the sign of every component of the vector; hence[8]

$$\hat{P}\mathbf{A}(\mathbf{k}) = -\mathbf{A}(-\mathbf{k}). \tag{6.1}$$

Although the decomposition of the photon's angular momentum into orbital and spin parts lacks physical meaning, it is nevertheless convenient to introduce "spin" s and "orbital" angular momentum l in a formal sense as auxiliary concepts describing the transformation properties of the wave function with respect to rotations: the value $s = 1$ reflects the vectorial character of the wave function, while l is the order of the spherical harmonics entering it. Here we have in mind the wave functions of states with definite values of the photon's angular momentum, which for a free particle take the form of spherical waves. The number l determines, in particular, the parity of the photon state, equal to

$$P = (-1)^{l+1} \tag{6.2}$$

In the same formal sense one can write the angular-momentum operator $\hat{\mathbf{j}}$ as the sum $\hat{\mathbf{s}} + \hat{\mathbf{l}}$. As is well known, the angular-momentum operator is associated with infinitesimally small rotations of the coordinate system; here, with the action of such a rotation on a vector field. Within the decomposition $\hat{\mathbf{s}} + \hat{\mathbf{l}}$, the operator $\hat{\mathbf{s}}$ acts on the vector index and intermixes the various components of the vector. The operator $\hat{\mathbf{l}}$ acts on these same components viewed as functions of momentum (or of coordinates). Let us count the number of states (at a given energy) that are possible for a given value j of the photon's angular momentum (setting aside the trivial $(2j + 1)$-fold degeneracy with respect to the direction of the angular momentum).

For independent $\mathbf{l}$ and $\mathbf{s}$ this counting is carried out simply by enumerating the ways in which, according to the rules of the vector model, the angular momenta $\hat{\mathbf{l}}$ and $\mathbf{s}$ can be composed so as to yield the required value of j. For a particle with spin $s = 1$ we would find in this way (for a given nonzero value of j) three states with the following values of l and parity:

$$l = j,\ P = (-1)^{l+1} = (-1)^{j+1}; \quad l = j \pm 1,\ P = (-1)^{l+1} = (-1)^{j}.$$

[8]We adopt the convention of defining the parity of a state by the action of the inversion operator on the polar vector $\mathbf{A}$ (or, equivalently, the corresponding electric vector $\mathbf{E} = i\omega\mathbf{A}$). This differs in sign from the action on the axial vector $\mathbf{H} = i[\mathbf{k}\mathbf{A}]$, since inversion does not reverse the direction of such a vector:

$$\hat{P}\mathbf{H}(\mathbf{k}) = \mathbf{H}(-\mathbf{k}).$$

If $j = 0$, there is only a single state (with $l = 1$) having parity $P = +1$.

This counting, however, has not accounted for the transversality condition on $\mathbf{A}$; all three components were regarded as independent. From the result obtained one must therefore subtract the number of states associated with a longitudinal vector. A longitudinal vector can be expressed as $\mathbf{k}\varphi(\mathbf{k})$, and it is apparent that, owing to its transformation behavior (with respect to rotations), its three components reduce to a single scalar φ [9]. The extra state that is excluded by transversality would therefore belong to a particle whose wave function is a scalar (a spinor of rank zero), that is, a particle with "spin 0" [10]. For this state the angular momentum j equals the order of the spherical harmonics appearing in φ. Its parity, viewed as a photon state, follows from how the inversion operator acts on the vector function $\mathbf{k}\varphi$:

$$\hat{P}(\mathbf{k}\varphi) = -(-\mathbf{k})\varphi(-\mathbf{k}) = (-1)^j\mathbf{k}\varphi(\mathbf{k}),$$

i.e. it equals $(-1)^j$. Thus, from the above count of states with parity $(-1)^j$ (two for $j \neq 0$ and one for $j = 0$) one must subtract one.

We arrive finally at the result that for nonzero photon angular momentum j there exist one even-parity and one odd-parity state. For $j = 0$ we obtain no states whatsoever. This means that the photon can never have zero angular momentum, so that j takes only the values $1, 2, 3, \ldots$ The impossibility of $j = 0$ is, moreover, obvious: the wave function of a state with vanishing angular momentum would have to be spherically symmetric, which is manifestly impossible for a transverse wave.

There is a standard nomenclature for the various photon states. When a photon carries angular momentum j with parity $(-1)^j$ it is called an *electric* 2^j*-pole* (or Ej) photon; when its parity is $(-1)^{j+1}$ it is called a *magnetic* 2^j*-pole* (or Mj) photon. In particular, the electric dipole photon has odd parity and $j = 1$; the electric quadrupole photon has even parity and $j = 2$; the magnetic dipole photon has even parity and $j = 1$ [11].

§ 7. Spherical waves of photons

Having established the admissible values of the photon's angular momentum, our task is now to find the wave functions that correspond to them [12].

Let us first consider a formal problem: to identify the vector functions that serve as eigenfunctions of the operators $\hat{\mathbf{j}}^2$ and $\hat{j}_z$; in doing so we do not stipulate in advance which particular functions enter the photon wave functions of interest, nor do we impose the transversality constraint.

[9] Indeed, when one speaks of the transformation character of a quantity under rotation, one means a transformation at a given point, i.e. at a given $\mathbf{k}$. Under such a transformation $\mathbf{k}\varphi(\mathbf{k})$ does not change at all, i.e. it behaves as a scalar.

[10] Let us emphasize once more that what is meant here is not the state of any real particle. The counting carried out here is purely formal and amounts, mathematically, to sorting the complete collection of quantities that mix under rotations into irreducible representations of the rotation group.

[11] This nomenclature mirrors the usage in the classical theory of radiation: as will be shown later (§§ 46–47), the production of electric- and magnetic-type photons is controlled by the respective electric and magnetic multipole moments of the system of charges.

[12] This problem was first studied by Heitler (W. Heitler, 1936). The form of the solution presented here is due to V. B. Berestetskii (1947).

The functions will be sought in the momentum representation, where the coordinate operator takes the form $\hat{\mathbf{r}} = i\partial/\partial\mathbf{k}$ (see III, (15.12)). For the orbital angular momentum operator one then obtains

$$\hat{\mathbf{l}} = [\hat{\mathbf{r}}\mathbf{k}] = -i\left[\mathbf{k}\frac{\partial}{\partial\mathbf{k}}\right],$$

i.e. it differs from the angular momentum operator in coordinate space only by the replacement of $\mathbf{r}$ with $\mathbf{k}$. Accordingly, the formal solution of the problem is identical in both representations.

Let us denote the desired eigenfunctions by $\mathbf{Y}_{jm}$ and call them *spherical vectors*. They must satisfy the equations

$$\hat{\mathbf{j}}^2 = j(j+1)\mathbf{Y}_{jm}, \qquad \hat{j}_z\mathbf{Y}_{jm} = m\mathbf{Y}_{jm} \tag{7.1}$$

(the z axis being a fixed direction in space). We shall demonstrate that any function of the form $\mathbf{a}Y_{jm}$ possesses this property, where $\mathbf{a}$ is a vector constructed from the unit vector $\mathbf{n} = \mathbf{k}/\omega$ and Y_{jm} are the ordinary (scalar) spherical harmonics. The latter will everywhere be defined in accordance with III, § 28:

$$Y_{lm}(\mathbf{n}) = (-1)^{\frac{m+|m|}{2}} i^l \sqrt{\frac{(2l+1)(l-|m|)!}{4\pi(l+|m|)!}}\, P_l^{|m|}(\cos\theta)\, e^{im\varphi} \tag{7.2}$$

(θ, φ being the spherical angles specifying the direction of $\mathbf{n}$) [13].

To show this, recall the commutation rule III, (29.4):

$$\{\hat{l}_i, a_k\}_- = ie_{ikl}a_l.$$

The right-hand side can be written as $(-\hat{s}_i a_k)$, where $\hat{\mathbf{s}}$ is the spin-1 operator (its action on a vector function is precisely $\hat{s}_i a_k = -ie_{ikl}a_l$; see III, § 57, problem 2). Consequently,

$$\hat{l}_i a_k - a_k\hat{l}_i = -\hat{s}_i a_k.$$

Making use of this relation, we find

$$\hat{j}_i a_k = (\hat{l}_i + \hat{s}_i)a_k = a_k\hat{l}_i.$$

It follows that

$$\hat{\mathbf{j}}^2(\mathbf{a}Y_{jm}) = \mathbf{a}\hat{l}^2 Y_{jm}, \qquad \hat{j}_z(\mathbf{a}Y_{jm}) = \mathbf{a}\hat{l}_z Y_{jm}.$$

Since the spherical harmonic Y_{jm} is an eigenfunction of $\hat{l}^2$ and $\hat{l}_z$ with eigenvalues $j(j+1)$ and m respectively, we arrive at equations (7.1).

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[13] For future reference, note the value of these functions at $\theta = 0$ ($\mathbf{n}$ along the z axis):

$$Y_{lm}(\mathbf{n}_z) = i^l\sqrt{\frac{2l+1}{4\pi}}\,\delta_{m0}. \tag{7.2a}$$

Three essentially distinct types of spherical vectors are obtained by choosing the vector $\mathbf{a}$ to be one of the following [14]:

$$\frac{\boldsymbol{\nabla}_{\mathbf{n}}}{\sqrt{j(j+1)}}, \qquad \frac{[\mathbf{n}\boldsymbol{\nabla}_{\mathbf{n}}]}{\sqrt{j(j+1)}}, \qquad \mathbf{n}. \tag{7.3}$$

The spherical vectors are thus defined as follows:

$$\begin{aligned}
\mathbf{Y}^{(\mathrm{e})}_{jm} &= \frac{1}{\sqrt{j(j+1)}}\,\boldsymbol{\nabla}_{\mathbf{n}} Y_{jm}, && P = (-1)^j;\\
\mathbf{Y}^{(\mathrm{m})}_{jm} &= [\mathbf{n}\mathbf{Y}^{(\mathrm{e})}_{jm}], && P = (-1)^{j+1};\\
\mathbf{Y}^{(\mathrm{l})}_{jm} &= \mathbf{n} Y_{jm}, && P = (-1)^j.
\end{aligned} \tag{7.4}$$

Next to each vector we have indicated its parity P. The spherical vectors of the three types are mutually orthogonal; $\mathbf{Y}^{(\mathrm{l})}_{jm}$ is longitudinal, while $\mathbf{Y}^{(\mathrm{e})}_{jm}$ and $\mathbf{Y}^{(\mathrm{m})}_{jm}$ are transverse with respect to $\mathbf{n}$.

The spherical vectors can be expressed through scalar spherical harmonics. In this expansion, $\mathbf{Y}^{(\mathrm{m})}_{jm}$ involves harmonics of a single order $l = j$ only, whereas $\mathbf{Y}^{(\mathrm{e})}_{jm}$ and $\mathbf{Y}^{(\mathrm{l})}_{jm}$ involve harmonics of orders $l = j \pm 1$. This is self-evident: one need only compare the parities listed in (7.4) with the parity $(-1)^{l+1}$ of a vector field, expressed via the order of the spherical harmonics it contains.

The spherical vectors within each type are mutually orthogonal and normalized according to

$$\int \mathbf{Y}_{jm}\mathbf{Y}^*_{j'm'}\,do = \delta_{jj'}\delta_{mm'}. \tag{7.5}$$

For the vectors $\mathbf{Y}^{(\mathrm{l})}_{jm}$ this follows immediately from the normalization of the spherical harmonics Y_{jm}. For $\mathbf{Y}^{(\mathrm{e})}_{jm}$ the normalization integral becomes

$$\frac{1}{j(j+1)}\int \boldsymbol{\nabla}_{\mathbf{n}} Y_{jm}\,\boldsymbol{\nabla}_{\mathbf{n}} Y^*_{j'm'}\,do = -\frac{1}{j(j+1)}\int Y^*_{j'm'}\,\Delta_{\mathbf{n}} Y_{jm}\,do,$$

and since $\Delta_{\mathbf{n}} Y_{jm} = -j(j+1)Y_{jm}$, we recover (7.5). The normalization of the vectors $\mathbf{Y}^{(\mathrm{m})}_{jm}$ reduces to an identical integral.

We remark that the spherical vectors (7.4) could also have been obtained without the direct verification of equations (7.1) carried out above — simply on the basis of general transformation-property arguments. Such reasoning led us in the preceding section to conclude that a vector function of the form $\mathbf{n}\varphi$ corresponds to a value of j equal to

[14] The operator $\boldsymbol{\nabla}_{\mathbf{n}} \equiv |\mathbf{k}|\boldsymbol{\nabla}_{\mathbf{k}}$ acts on functions that depend only on the direction of $\mathbf{n}$. In spherical coordinates it has just two components:

$$\boldsymbol{\nabla}_{\mathbf{n}} = \left(\frac{\partial}{\partial\theta}, \frac{1}{\sin\theta}\frac{\partial}{\partial\varphi}\right).$$

The operator denoted below by $\Delta_{\mathbf{n}}$ is the angular part of the Laplacian:

$$\Delta_{\mathbf{n}} = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}.$$

the order of the spherical harmonics appearing in φ; setting $\varphi = Y_{jm}$ further fixes the projection m. In this way one immediately arrives at the spherical vectors $\mathbf{Y}^{(1)}_{jm}$. The transformation-property arguments of § 6, however, remain valid if the factor $\mathbf{n}$ in the product $\mathbf{n}\varphi$ is replaced by $\boldsymbol{\nabla}_{\mathbf{n}}$ or by $[\mathbf{n}\boldsymbol{\nabla}_{\mathbf{n}}]$, yielding in this manner the spherical vectors of the other two types.

Let us return to the photon wave functions. For an electric-type photon (Ej) the vector $\mathbf{A}(\mathbf{k})$ must have parity $(-1)^j$. Among the spherical vectors, $\mathbf{Y}^{(e)}_{jm}$ and $\mathbf{Y}^{(1)}_{jm}$ possess this parity; of the two, however, only the former satisfies the transversality condition. For a magnetic-type photon (Mj) the vector $\mathbf{A}(\mathbf{k})$ must have parity $(-1)^{j+1}$; only the spherical vector $\mathbf{Y}^{(m)}_{jm}$ has this parity. The wave functions of a photon with definite angular momentum j, projection m, and energy ω are therefore

$$\mathbf{A}_{\omega jm}(\mathbf{k}) = \frac{4\pi^2}{\omega^{3/2}}\,\delta(|\mathbf{k}| - \omega)\mathbf{Y}_{jm}(\mathbf{n}), \tag{7.6}$$

where one should write $\mathbf{Y}^{(e)}_{jm}$ or $\mathbf{Y}^{(m)}_{jm}$ for $\mathbf{Y}_{jm}$ according as the photon is of electric or magnetic type; the specified energy is incorporated through the factor $\delta(|\mathbf{k}| - \omega)$. The functions (7.6) are normalized by the condition

$$\frac{1}{(2\pi)^4}\int \omega\omega' \mathbf{A}^*_{\omega' j'm'}(\mathbf{k})\mathbf{A}_{\omega jm}(\mathbf{k})\,d^3k = \omega\delta(\omega' - \omega)\delta_{jj'}\delta_{mm'}. \tag{7.7}$$

For the coordinate-representation wave functions, condition (7.7) is equivalent to [15]

$$\frac{1}{4\pi}\int \left\{\mathbf{E}^*_{\omega' j'm'}(\mathbf{r})\mathbf{E}_{\omega jm}(\mathbf{r}) + \mathbf{H}^*_{\omega' j'm'}(\mathbf{r})\mathbf{H}_{\omega jm}(\mathbf{r})\right\} d^3x = \omega\delta(\omega' - \omega)\delta_{jj'}\delta_{mm'}. \tag{7.8}$$

Indeed, when written in terms of potentials, the integral on the left-hand side takes the form

$$\frac{1}{2\pi}\int \mathbf{A}^*_{\omega' j'm'}(\mathbf{r})\mathbf{A}_{\omega jm}(\mathbf{r})\,\omega'\omega\,d^3x.$$

One must substitute here the Fourier expansions

$$\begin{aligned}\mathbf{A}_{\omega jm}(\mathbf{r}) &= \int \mathbf{A}_{\omega jm}(\mathbf{k})\,e^{i\mathbf{kr}}\,\frac{d^3k}{(2\pi)^3},\\ \mathbf{A}^*_{\omega' j'm'}(\mathbf{r}) &= \int \mathbf{A}^*_{\omega' j'm'}(\mathbf{k}')\,e^{-i\mathbf{k}'\mathbf{r}}\,\frac{d^3k'}{(2\pi)^3}.\end{aligned} \tag{7.9}$$

After that, integration over d^3x produces the delta function $(2\pi)^3\delta(\mathbf{k}' - \mathbf{k})$, which is removed by the integration over d^3k', bringing the integral to the form (7.7).

Up to this point we have assumed the transverse gauge for the potentials, in which the scalar potential $\Phi = 0$. In various applications, however, other gauge choices for a spherical wave may prove more convenient. An admissible gauge transformation of the potentials in the momentum representation consists of the substitution

$$\mathbf{A} \to \mathbf{A} + \mathbf{n}f(\mathbf{k}), \qquad \Phi \to \Phi + f(\mathbf{k}),$$

[15] This condition is of the same type as (2.22). The appearance of the factor $\delta(\omega' - \omega)$ on the right-hand side is due to the fact that the field (a spherical wave) is being considered in all of infinite space rather than in a finite volume $V = 1$.

with $f(\mathbf{k})$ an arbitrary function. We choose it here so that the new potentials are still expressed through the same spherical harmonics and continue to have definite parity. For an electric-type photon these requirements restrict the potentials to the following form:

$$\begin{aligned} \mathbf{A}^{(\mathrm{e})}_{\omega jm}(\mathbf{k}) &= \frac{4\pi^2}{\omega^{3/2}}\,\delta(|\mathbf{k}|-\omega)\big(\mathbf{Y}^{(\mathrm{e})}_{jm}+C\mathbf{n}Y_{jm}\big),\\ \Phi^{(\mathrm{e})}_{\omega jm}(\mathbf{k}) &= \frac{4\pi^2}{\omega^{3/2}}\,\delta(|\mathbf{k}|-\omega)\,CY_{jm}, \end{aligned} \tag{7.10}$$

where C is an arbitrary constant. For a magnetic-type photon, on the other hand, such an addition to $\mathbf{A}^{(\mathrm{m})}(\mathbf{k})$ would destroy its definite parity, so under the same conditions the choice (7.6) is uniquely fixed.

The probability that a photon of given angular momentum and parity is detected traveling in the direction $\mathbf{n}$ within the solid-angle element do equals, by (3.5) and (7.6),

$$w(\mathbf{n})do = \left|\mathbf{Y}^{(\mathrm{e})}_{jm}\right|^2 do. \tag{7.11}$$

We have written the expression for an E-type photon. Since, however, $\left|\mathbf{Y}^{(\mathrm{m})}_{jm}\right|^2 = \left|\mathbf{Y}^{(\mathrm{e})}_{jm}\right|^2$, the probability distributions $w(\mathbf{n})$ for photons of both types are identical.

The squared modulus $\left|\mathbf{Y}^{(\mathrm{e})}_{jm}\right|^2$ is independent of the azimuthal angle φ (the factors $e^{\pm im\varphi}$ in the spherical harmonics cancel). The probability distribution $w(\mathbf{n})$ is therefore axially symmetric about the z axis. Furthermore, since each spherical vector has definite parity, the squared moduli are even under inversion, i.e. under the replacement $\theta \to \pi - \theta$; this means that the function $w(\theta)$, when expanded in Legendre polynomials, contains only polynomials of even order. The determination of the expansion coefficients reduces to computing integrals of products of three spherical harmonics followed by summation over components. Both steps are carried out with the aid of formulas derived in III, §§ 107–108, and lead to the following result:

$$\begin{aligned} w(\theta) = (-1)^{m+1}\frac{(2j+1)^2}{4\pi}\sum_{n=0}^{\infty}(4n+1)\begin{pmatrix} j & j & 2n\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix} j & j & 2n\\ m & -m & 0\end{pmatrix}\times\\ \times\begin{Bmatrix} j & j & 1\\ j & j & 2n\end{Bmatrix}P_{2n}(\cos\theta). \end{aligned} \tag{7.12}$$

Finally, we present expressions for the components of the spherical vectors as expansions in spherical harmonics. We employ the "spherical components" of a vector $\mathbf{f}$, defined in accordance with III, § 107; the components f_λ of the vector $\mathbf{f}$ are

$$f_0 = if_z, \quad f_{+1} = -\frac{i}{\sqrt{2}}(f_x + if_y), \quad f_{-1} = \frac{i}{\sqrt{2}}(f_x - if_y). \tag{7.13}$$

Introducing "circular basis vectors":

$$\mathbf{e}^{(0)} = i\mathbf{e}^{(z)}, \quad \mathbf{e}^{(+1)} = -\frac{i}{\sqrt{2}}\big(\mathbf{e}^{(x)} + i\mathbf{e}^{(y)}\big), \quad \mathbf{e}^{(-1)} = \frac{i}{\sqrt{2}}\big(\mathbf{e}^{(x)} - i\mathbf{e}^{(y)}\big) \tag{7.14}$$

($\mathbf{e}^{(x,y,z)}$ being the unit vectors along the x, y, z axes), we have

$$\mathbf{f} = \sum_\lambda (-1)^{1-\lambda} f_{-\lambda} \mathbf{e}^{(\lambda)}, \qquad f_\lambda = (-1)^{1-\lambda} \mathbf{f}\mathbf{e}^{(-\lambda)*} = \mathbf{f}\mathbf{e}^{(\lambda)}. \tag{7.15}$$

The spherical components of the spherical vectors are expressed through $3j$-symbols and spherical harmonics by the following formulas:

$$\begin{aligned}
(-1)^{j+m+\lambda+1}\big(\mathbf{Y}^{(\mathrm{e})}_{jm}\big)_\lambda &= -\sqrt{j}\begin{pmatrix} j+1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j+1,m+\lambda} \\
&\quad + \sqrt{j+1}\begin{pmatrix} j-1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j-1,m+\lambda}, \\
(-1)^{j+m+\lambda+1}\big(\mathbf{Y}^{(\mathrm{m})}_{jm}\big)_\lambda &= -\sqrt{2j+1}\begin{pmatrix} j & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j,m+\lambda}, \\
(-1)^{j+m+\lambda+1}\big(\mathbf{Y}^{(\mathrm{l})}_{jm}\big)_\lambda &= \sqrt{j+1}\begin{pmatrix} j+1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j+1,m+\lambda} \\
&\quad + \sqrt{j}\begin{pmatrix} j-1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j-1,m+\lambda}.
\end{aligned} \tag{7.16}$$

These formulas are derived as follows. Each of the three spherical vectors has the form $\mathbf{Y}_{jm} = \mathbf{a}Y_{jm}$, where $\mathbf{a}$ is one of the three vectors (7.3). Therefore

$$\mathbf{Y}_{jm} = \sum_{lm'} \langle lm'|\mathbf{a}|jm\rangle Y_{lm'},$$

so that one must determine the matrix elements of the vectors $\mathbf{a}$ between eigenstates of the orbital angular momentum. From formula (107.6) of III it follows that

$$\langle lm'|a_\lambda|jm\rangle = i(-1)^{j_{\max}-m'}\begin{pmatrix} l & 1 & j \\ -m' & \lambda & m \end{pmatrix}\langle l\|a\|j\rangle,$$

where $j_{\max}$ is the larger of the numbers l and j. It therefore suffices to know the nonvanishing reduced matrix elements $\langle l\|a\|j\rangle$. The relevant formulas are:

$$\begin{aligned}
\langle l-1\|n\|l\rangle &= \langle l\|n\|l-1\rangle^* = i\sqrt{l}, \\
\langle l\|\nabla_{\mathbf{n}}\|l-1\rangle &= i(l-1)\sqrt{l}, \\
\langle l-1\|\nabla_{\mathbf{n}}\|l\rangle &= i(l+1)\sqrt{l}, \\
\langle l\|[\mathbf{n}\nabla_{\mathbf{n}}]\|l\rangle &= i\sqrt{l(l+1)(2l+1)}.
\end{aligned} \tag{7.17}$$

§ 8. Polarization of the photon

The polarization vector $\mathbf{e}$ plays for the photon the role of the "spin part" of the wave function (with the reservations expressed in § 6 regarding the notion of photon spin).

The various cases arising for the polarization of a photon are in no way distinct from the possible types of polarization of a classical electromagnetic wave (see II, § 48).

An arbitrary polarization $\mathbf{e}$ can be represented as a superposition of two mutually orthogonal polarizations $\mathbf{e}^{(1)}$ and $\mathbf{e}^{(2)}$ ($\mathbf{e}^{(1)}\mathbf{e}^{(2)*} = 0$), chosen in some definite manner. In the decomposition

$$\mathbf{e} = e_1\mathbf{e}^{(1)} + e_2\mathbf{e}^{(2)} \tag{8.1}$$

the squared moduli of the coefficients e_1 and e_2 give the probability that the photon has polarization $\mathbf{e}^{(1)}$ or $\mathbf{e}^{(2)}$.

As these two basis polarizations one may choose two mutually perpendicular linear polarizations. Alternatively, any polarization can be decomposed into two circular ones with opposite senses of rotation. We shall denote the right and left circular polarization vectors by $\mathbf{e}^{(+1)}$ and $\mathbf{e}^{(-1)}$ respectively; in a coordinate system $\xi\eta\zeta$ with the ζ axis along the photon's direction $\mathbf{n} = \mathbf{k}/\omega$,

$$\mathbf{e}^{(+1)} = -\frac{i}{\sqrt{2}}\left(\mathbf{e}^{(\xi)} + i\mathbf{e}^{(\eta)}\right), \qquad \mathbf{e}^{(-1)} = \frac{i}{\sqrt{2}}\left(\mathbf{e}^{(\xi)} - i\mathbf{e}^{(\eta)}\right). \tag{8.2}$$

The existence of two distinct polarizations of the photon (at a given momentum) means, in other words, that every eigenvalue of the momentum is doubly degenerate. This circumstance is intimately connected with the fact that the photon mass equals zero.

For a freely moving particle of nonvanishing mass a rest frame always exists. It is clear that the intrinsic symmetry properties of the particle as such manifest themselves precisely in this frame. The symmetry to be considered is with respect to all possible rotations about the center, that is, with respect to the full spherical symmetry group. The quantity characterizing the symmetry properties of the particle under this group is its spin s, which determines the multiplicity of the degeneracy (the number $2s + 1$ of distinct wave functions that transform into one another). In particular, a particle whose wave function is a vector (three components) has spin 1.

For a particle of zero mass, however, no rest frame exists — in every reference frame it travels at the speed of light. For such a particle there is always a preferred spatial direction — the direction of the momentum vector $\mathbf{k}$ (the ζ axis). In this situation the full three-dimensional rotation symmetry is evidently absent, and one may speak only of axial symmetry about the distinguished axis.

Under axial symmetry only the *helicity* of the particle is conserved — the projection of angular momentum onto the ζ axis; we denote it by λ [16]. If one further demands symmetry under reflections in planes passing through the ζ axis, then states differing in the sign of λ are mutually degenerate; for $\lambda \neq 0$ there is therefore a twofold degeneracy [17]. The state of a photon with a definite momentum belongs to precisely this type of twofold-degenerate state. It is described by a "spin" wave function that is a vector $\mathbf{e}$ in the $\xi\eta$ plane; the two components of this vector transform into each other under all rotations about the ζ axis and under reflections in planes containing that axis.

The various polarization cases of a photon are in a definite correspondence with the possible values of its helicity. This correspondence can be established using formulas III,

[16] In contrast to the projection m of the angular momentum onto a fixed direction (the z axis) in space, discussed in the preceding section.

[17] We note that the electronic terms of a diatomic molecule are classified in exactly the same way (see III, § 78).

(57.9), which relate the components of a vector wave function to the components of the equivalent rank-2 spinor[18]. The projections $\lambda = +1$ and -1 correspond to vectors $\mathbf{e}$ that have only the component $e_\xi - ie_\eta$ or $e_\xi + ie_\eta$ nonzero, i.e. $\mathbf{e} = \mathbf{e}^{(+1)}$ or $\mathbf{e} = \mathbf{e}^{(-1)}$ respectively. In other words, $\lambda = +1$ and -1 correspond to the right and left circular polarizations of the photon (in § 16 the same result will be derived by directly computing the eigenfunctions of the spin-projection operator). Thus the projection of the photon's angular momentum onto the direction of its motion can take only the two values (± 1); the value 0 is not possible.

A photon state with definite momentum and polarization is a pure state (in the sense explained in III, § 14); it is described by a wave function and constitutes a complete quantum-mechanical specification of the particle (photon) state. There also exist "mixed" photon states corresponding to a less complete description, carried out not by a wave function but only by a density matrix.

Let us examine a photon state that is mixed in its polarization but possesses a definite momentum $\mathbf{k}$. Such a state (called a state of *partial polarization*) admits a "coordinate" wave function.

The polarization density matrix of a photon is a second-rank tensor $\rho_{\alpha\beta}$ defined in the plane orthogonal to $\mathbf{n}$ (the $\xi\eta$ plane; the indices α, β assume only two values). This tensor is Hermitian:

$$\rho_{\alpha\beta} = \rho^*_{\beta\alpha}, \tag{8.3}$$

and normalized by the condition

$$\rho_{\alpha\alpha} = \rho_{11} + \rho_{22} = 1. \tag{8.4}$$

By virtue of (8.3) the diagonal components ρ_{11} and ρ_{22} are real, and one determines the other through condition (8.4). The component ρ_{12} is complex, while $\rho_{21} = \rho^*_{12}$. Altogether, then, the density matrix is characterized by three real parameters.

Once the polarization density matrix is known, one can find the probability that the photon has any given definite polarization $\mathbf{e}$. This probability is obtained by "projecting" the tensor $\rho_{\alpha\beta}$ onto the direction of the vector $\mathbf{e}$, i.e. it equals the quantity

$$\rho_{\alpha\beta} e^*_\alpha e_\beta. \tag{8.5}$$

Thus the components ρ_{11} and ρ_{22} are the probabilities of linear polarizations along the ξ and η axes. Projection onto the vectors (8.2) yields the probabilities of the two circular polarizations:

$$\tfrac{1}{2}\left[1 \pm i(\rho_{12} - \rho_{21})\right]. \tag{8.6}$$

In both form and substance the properties of the tensor $\rho_{\alpha\beta}$ coincide with those of the tensor $J_{\alpha\beta}$ that describes partially polarized light in the classical theory (see II, § 50). Let us recall some of these properties here.

For a pure state of definite polarization $\mathbf{e}$, the tensor $\rho_{\alpha\beta}$ reduces to products of the components of the vector $\mathbf{e}$:

$$\rho_{\alpha\beta} = e_\alpha e^*_\beta. \tag{8.7}$$

[18]Recall that the wave-function components, viewed as probability amplitudes for the various values of the angular-momentum projection of the particle (which is what is meant here), correspond to the contravariant components of the spinor.

In this case the determinant $|\rho_{\alpha\beta}| = 0$. In the opposite case of an unpolarized photon, all polarization directions are equally probable, i.e.

$$\rho_{\alpha\beta} = \delta_{\alpha\beta}/2, \tag{8.8}$$

and then $|\rho_{\alpha\beta}| = 1/4$.

For the general case of partial polarization it is convenient to introduce three real Stokes parameters ξ_1, ξ_2, ξ_3 [19]; the density matrix is then expressible as

$$\rho_{\alpha\beta} = \frac{1}{2}\begin{pmatrix} 1+\xi_3 & \xi_1 - i\xi_2 \\ \xi_1 + i\xi_2 & 1-\xi_3 \end{pmatrix}. \tag{8.9}$$

All three parameters range between -1 and $+1$. In the unpolarized state $\xi_1 = \xi_2 = \xi_3 = 0$; for a completely polarized photon $\xi_1^2 + \xi_2^2 + \xi_3^2 = 1$.

The parameter ξ_3 characterizes linear polarization along the ξ or η axis; the probability of linear polarization along these axes is $(1+\xi_3)/2$ and $(1-\xi_3)/2$ respectively. The values $\xi_3 = +1$ or -1 therefore correspond to complete polarization in these directions.

The parameter ξ_1 characterizes linear polarization along directions making an angle $\varphi = \pi/4$ or $\varphi = -\pi/4$ with the ξ axis. The probability of linear polarization in these directions is $(1+\xi_1)/2$ or $(1-\xi_1)/2$; this is easily verified by projecting the tensor $\rho_{\alpha\beta}$ onto the directions $\mathbf{e} = (1, \pm 1)/\sqrt{2}$.

Lastly, the parameter ξ_2 represents the degree of circular polarization; by (8.6), the probability of right or left circular polarization is $(1+\xi_2)/2$ or $(1-\xi_2)/2$ respectively. Because these two polarizations answer to helicities $\lambda = \pm 1$, it is evident that in general ξ_2 coincides with the mean helicity of the photon. Note further that for a pure state with polarization $\mathbf{e}$,

$$\xi_2 = i[\mathbf{e}\mathbf{e}^*]\mathbf{n}. \tag{8.10}$$

Recall (see II, § 50) that the quantities ξ_2 and $\sqrt{\xi_1^2 + \xi_3^2}$ are invariant under Lorentz transformations.

In what follows we shall also need to know how the Stokes parameters transform under time reversal. One readily verifies that they are invariant under this operation. Since the result clearly does not depend on whether the polarization state is pure or mixed, it is enough to check invariance for a pure state. In quantum mechanics, time reversal replaces the wave function with its complex conjugate (see III, § 18). For a plane-polarized wave the corresponding substitution is [20]

$$\mathbf{k} \to -\mathbf{k}, \qquad \mathbf{e} \to -\mathbf{e}^*. \tag{8.11}$$

Under this transformation the symmetric part of the density matrix, $(1/2)(e_\alpha e_\beta^* + e_\beta e_\alpha^*)$, and hence the parameters ξ_1 and ξ_3 remain unchanged. The invariance of ξ_2 under the

[19]The notation for the parameters should not be confused with that for the ξ axis!

[20]The extra sign reversal of $\mathbf{e}$ is due to the fact that time reversal changes the sign of the electromagnetic vector potential. The scalar potential is unaffected; consequently, for the 4-vector e time reversal takes the form

$$(e_0, \mathbf{e}) \to (e_0^*, -\mathbf{e}^*). \tag{8.11a}$$

same transformation is apparent from (8.10); it is also obvious already from the physical meaning of ξ_2 as the mean helicity. Indeed, helicity is the projection of the angular momentum $\mathbf{j}$ onto the direction $\mathbf{n}$, that is, the product $\mathbf{jn}$; time reversal reverses the sign of both these vectors.

For the calculations that follow, the photon density matrix must be cast in four-dimensional form, that is, expressed as a certain 4-tensor $\rho_{\mu\nu}$. When the photon is polarized and described by a 4-vector e_μ, a natural definition of this tensor is

$$\rho_{\mu\nu} = e_\mu e_\nu^*. \tag{8.12}$$

In the three-dimensionally transverse gauge, $e = (0, \mathbf{e})$, if one of the spatial coordinate axes is chosen along $\mathbf{n}$, the nonvanishing components of this 4-tensor coincide with (8.7).

For an unpolarized photon the three-dimensionally transverse gauge yields a tensor $\rho_{\mu\nu}$ with components

$$\rho_{ik} = \tfrac{1}{2}(\delta_{ik} - n_i n_k), \qquad \rho_{0i} = \rho_{i0} = \rho_{00} = 0 \tag{8.13}$$

(if one of the axes coincides with the direction of $\mathbf{n}$, we return to (8.8)). Using the tensor $\rho_{\mu\nu}$ directly in this three-dimensional form is, however, inconvenient. We may instead perform a gauge transformation; for the density matrix this is a transformation of the type

$$\rho_{\mu\nu} \to \rho_{\mu\nu} + \chi_\mu k_\nu + \chi_\nu k_\mu, \tag{8.14}$$

where χ_μ are arbitrary functions. Setting

$$\chi_0 = -1/(4\omega), \qquad \chi_i = k_i/(4\omega^2)$$

we obtain in place of (8.13) the simple four-dimensional expression

$$\rho_{\mu\nu} = -g_{\mu\nu}/2. \tag{8.15}$$

The four-dimensional representation of the density matrix of a partially polarized photon is readily obtained by first rewriting the two-dimensional tensor (8.9) in three-dimensional form. In the three-dimensionally transverse gauge: $e^{(1)} = (0, \mathbf{e}^{(1)})$, $e^{(2)} = (0, \mathbf{e}^{(2)})$. The four-dimensional photon density matrix is therefore

$$\begin{aligned}\rho_{\mu\nu} = {} & (1/2)(e_\mu^{(1)} e_\nu^{(1)} + e_\mu^{(2)} e_\nu^{(2)}) + (\xi_1/2)(e_\mu^{(1)} e_\nu^{(2)} + e_\mu^{(2)} e_\nu^{(1)}) - \\ & - (i\xi_2/2)(e_\mu^{(1)} e_\nu^{(2)} - e_\mu^{(2)} e_\nu^{(1)}) + (\xi_3/2)(e_\mu^{(1)} e_\nu^{(1)} - e_\mu^{(2)} e_\nu^{(2)}).\end{aligned} \tag{8.17}$$

The convenience of any particular choice of the 4-vectors $e^{(1)}$, $e^{(2)}$ depends on the specific conditions of the problem at hand.

It should be borne in mind that the conditions (8.16) do not fix the choice of $e^{(1)}$ and $e^{(2)}$ uniquely. If any 4-vector e_μ satisfies these conditions, so does any 4-vector of the form $e_\mu + \chi k_\mu$ (since $k^2 = 0$). This ambiguity is tied to the gauge freedom of the density matrix.

The first term in (8.17) corresponds to the unpolarized state. It could therefore be replaced, by (8.15), with $-g_{\mu\nu}/2$. Such a replacement is again equivalent to a gauge transformation. When working with 4-tensors of the form (8.17), decomposed in two independent 4-vectors, it is useful to adopt the following formal device. Writing (8.17) as

$$\rho_{\mu\nu} = \sum_{a,b=1}^{2} \rho^{(ab)} e_\mu^{(a)} e_\nu^{(b)},$$

we represent the coefficients $\rho^{(ab)}$ as a 2×2 matrix

$$\rho = \begin{pmatrix} \rho^{(11)} & \rho^{(12)} \\ \rho^{(21)} & \rho^{(22)} \end{pmatrix}.$$

Any Hermitian 2×2 matrix can be expanded over four linearly independent 2×2 matrices — the Pauli matrices σ_x, σ_y, σ_z and the identity matrix 1:

$$\rho = (1/2)(1 + \boldsymbol{\xi}\boldsymbol{\sigma}), \qquad \boldsymbol{\xi} = (\xi_1, \xi_2, \xi_3), \tag{8.18}$$

as is easily verified by direct comparison with (8.17) using the well-known explicit forms of the Pauli matrices (18.5) (grouping the three quantities ξ_1, ξ_2, ξ_3 into a "vector" $\boldsymbol{\xi}$ is of course purely formal and serves only to shorten the notation).

The requisite generalization is achieved by replacing these 3-vectors by spacelike real 4-vectors $e^{(1)}$, $e^{(2)}$, orthogonal to each other and to the photon 4-momentum k:

$$e^{(1)2} = e^{(2)2} = -1, \qquad e^{(1)}e^{(2)} = 0, \qquad e^{(1)}k = e^{(2)}k = 0. \tag{8.16}$$

Problem

Write the photon density matrix in the representation whose "coordinate axes" are the circular basis vectors (8.2).

Solution. The components of the tensor $\rho'_{\alpha\beta}$ in the new axes ($\alpha, \beta = \pm 1$) are obtained by projecting the tensor (8.9) onto the basis vectors (8.2):

$$\rho'_{11} = \rho_{\alpha\beta} e_\alpha^{(+1)*} e_\beta^{(+1)}, \qquad \rho'_{1,-1} = \rho_{\alpha\beta} e_\alpha^{(+1)*} e_\beta^{(-1)}, \quad \ldots,$$

$$\rho' = \frac{1}{2}\begin{pmatrix} 1 + \xi_2 & -\xi_3 + i\xi_1 \\ -\xi_3 - i\xi_1 & 1 - \xi_2 \end{pmatrix}.$$

§ 9. The two-photon system

Arguments analogous to those developed in § 6 make it possible to count the number of allowed states in the more involved case of a system of two photons as well (*L. Landau*, 1948).

Let us work in the center-of-inertia frame of the two photons, where $\mathbf{k}_1 = -\mathbf{k}_2 \equiv \mathbf{k}$[21]. In the momentum representation the wave function of the pair can be cast as a three-dimensional rank-2 tensor $A_{ik}(\mathbf{n})$ built bilinearly from the components of the individual

[21] A center-of-inertia frame can be found in every case except when the two photons propagate in the same direction. For such a pair the total momentum $\mathbf{k}_1 + \mathbf{k}_2$ and total energy $\omega_1 + \omega_2$ are connected by exactly the same relation as for one photon, so no frame with $\mathbf{k}_1 + \mathbf{k}_2 = 0$ exists.

photon vector wave functions; each of its two indices refers to one photon (**n** denotes the unit vector in the direction of **k**). The transversality requirement for each photon is encoded in the orthogonality of A_{ik} to **n**:

$$A_{il}n_l = 0, \qquad A_{lk}n_l = 0. \tag{9.1}$$

Interchanging the two photons amounts to transposing the indices of the tensor A_{ik} together with a simultaneous reversal of the sign of **n**. Because photons obey Bose statistics,

$$A_{ik}(-\mathbf{n}) = A_{ki}(\mathbf{n}). \tag{9.2}$$

The tensor A_{ik} is, in general, not symmetric in its indices. We decompose it into its symmetric part (s_{ik}) and antisymmetric part (a_{ik}): $A_{ik} = s_{ik} + a_{ik}$. Each of these parts must individually satisfy the relation (9.2) (as well as the orthogonality conditions (9.1)). This gives

$$s_{ik}(-\mathbf{n}) = s_{ik}(\mathbf{n}), \tag{9.3}$$

$$a_{ik}(-\mathbf{n}) = -a_{ik}(\mathbf{n}). \tag{9.4}$$

Spatial inversion does not by itself reverse the sign of the components of a second-rank tensor, but it does reverse the sign of **n**. From (9.3) it is therefore apparent that the wave function s_{ik} is symmetric under inversion, i.e. it corresponds to even-parity states of the photon system; the wave function a_{ik}, on the other hand, corresponds to odd-parity states.

An antisymmetric second-rank tensor is equivalent (dual) to an axial vector **a** whose components are expressed via the tensor components as $a_i = \frac{1}{2}e_{ikl}a_{kl}$, where e_{ikl} is the antisymmetric unit tensor (see II, § 6). The orthogonality of the tensor a_{kl} to the vector **n** implies that the vectors **a** and **n** are parallel[22]. One may therefore write $\mathbf{a} = \mathbf{n}\varphi(\mathbf{n})$, where φ is a scalar; by (9.4) we must have $\mathbf{a}(-\mathbf{n}) = -\mathbf{a}(\mathbf{n})$, and hence

$$\varphi(-\mathbf{n}) = \varphi(\mathbf{n}).$$

This relation means that the scalar φ can be built linearly only from spherical harmonics of even order L (including zero).

We see that the antisymmetric tensor a_{ik}, by its transformation properties (under rotations), is equivalent to a single scalar (cf. the footnote on p. 34). Assigning to the latter a "spin" of 0, we find that the angular momentum of the state is $J = L$. Thus the tensor a_{ik} corresponds to odd-parity states of the photon system with even total angular momentum J.

Let us turn to the symmetric tensor s_{ik}. Since it is even under reversal of the sign of **n**, it describes even-parity states of the photon system. It follows likewise that all components of s_{ik} are expressed through spherical harmonics of even order L (including $L = 0$). An arbitrary symmetric second-rank tensor can be decomposed, as is well known, into a scalar (s_{ii}) and a traceless symmetric tensor (s'_{ik}) with $s'_{ii} = 0$.

[22]We have: $a_{ik} = e_{ikl}a_l$, and the orthogonality condition yields

$$a_{ik}n_k = e_{ikl}a_l n_k = [\mathbf{na}]_i = 0.$$

The scalar s_{ii} is assigned "spin" 0, so the angular momentum of the corresponding states is $J = L$, which is even. The tensor s'_{ik} corresponds to "spin" 2 (see III, § 57). Composing this "spin" with the even "orbital angular momentum" L according to the addition rule for angular momenta, we find that for a given even $J \neq 0$ there are three states (with $L = J \pm 2, J$), while for odd $J \neq 1$ there are two states (with $L = J \pm 1$). The exceptions are $J = 0$ with a single state ($L = 2$) and $J = 1$ with a single state ($L = 2$).

In this counting, however, we have not yet accounted for the orthogonality of the tensor s_{ik} to the vector $\mathbf{n}$. One must therefore subtract from the number of states obtained those corresponding to a symmetric second-rank tensor that is "parallel" to the vector $\mathbf{n}$. Such a tensor (denote it s''_{ik}) can be written as

$$s''_{ik} = n_i b_k + n_k b_i,$$

where $\mathbf{b}$ is some vector. By (9.3) this vector must satisfy $\mathbf{b}(-\mathbf{n}) = -\mathbf{b}(\mathbf{n})$. The tensor s''_{ik} responsible for the "extra" states is thus equivalent to an odd-parity vector. This vector must consequently be expressed through spherical harmonics of odd orders L only. Noting further that a vector corresponds to "spin" 1, we find that for each even $J \neq 0$ there are two states (with $L = J \pm 1$), and for each odd J a single state (with $L = J$); the special case $J = 0$ has one state (with $L = 1$).

Collecting all the results obtained, we arrive at the following table giving the number of possible even- and odd-parity states of a system of two photons (with vanishing total momentum) for the various values of the total angular momentum J:

J	Even	Odd
0	1	1
1	–	–
$2k$	2	1
$2k+1$	1	–

(9.5)

(k being a positive integer other than zero). We observe that for odd J no odd-parity states exist, while the value $J = 1$ is altogether excluded[23].

The wave function of the two-photon system A_{ik} determines the correlation between their polarizations. The probability that both photons simultaneously possess definite polarizations $\mathbf{e}_1$ and $\mathbf{e}_2$ is proportional to

$$A_{ik} e^*_{1i} e^*_{2k}.$$

In other words, if the polarization $\mathbf{e}_1$ of one photon is specified, then the polarization of the second photon, $\mathbf{e}_2$, is proportional to

$$e_{2k} \propto A_{ik} e^*_{1i}. \tag{9.6}$$

In odd-parity states of the system, A_{ik} coincides with the antisymmetric tensor a_{ik}. In that case

$$\mathbf{e}_2 \mathbf{e}^*_1 \propto a_{ik} e^*_{1i} e^*_{1k} = 0,$$

[23] For an alternative derivation of these results, see problem 1 to § 69.

i.e. the polarizations of the two photons are mutually orthogonal. For linear polarization this signifies that their directions are perpendicular, while for circular polarization it means opposite senses of rotation.

The even-parity state with $J = 0$ is described by a symmetric tensor reducing to a scalar,

$$s_{ik} = \text{const} \cdot (\delta_{ik} - n_i n_k).$$

From (9.6) we then obtain $\mathbf{e}_1 = \mathbf{e}_2^*$. For linear polarization this means that their directions are parallel, whereas for circular polarization it means opposite senses of rotation. The latter fact is self-evident: when $J = 0$ the sum of the angular-momentum projections of the two photons onto any common direction $\mathbf{k}$ must in all cases vanish (the projections onto the opposite directions $\mathbf{k}_1$ and $\mathbf{k}_2$, i.e. the helicities, are accordingly equal).

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